

# Boosting efficiency in state estimation of power systems by leveraging attention mechanism

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## ABSTRACT

Ensuring stability and reliability in power systems requires accurate state estimation, which is challenging due to the growing network size, noisy measurements, and nonlinear power-flow equations. In this paper, we introduce the Graph Attention Estimation Network (GAEN) model to tackle power system state estimation (PSSE) by capitalizing on the inherent graph structure of power grids. This approach facilitates efficient information exchange among interconnected buses, yielding a distributed, computationally efficient architecture that is also resilient to cyber-attacks. We develop a thorough approach by utilizing Graph Convolutional Neural Networks (GCNNs) and attention mechanism in PSSE based on Supervisory Control and Data Acquisition (SCADA) and Phasor Measurement Unit (PMU) measurements, addressing the limitations of previous learning architectures. In accordance with the empirical results obtained from the experiments, the proposed method demonstrates superior performance and scalability compared to existing techniques. Furthermore, the amalgamation of local topological configurations with nodal-level data yields a heightened efficacy in the domain of state estimation. This work marks a significant achievement in the design of advanced learning architectures in PSSE, contributing and fostering the development of more reliable and secure power system operations.

## 1. Introduction

In the past few years, our reliance on power grids has been dramatically underscored by catastrophic failures leading to immense societal and economic costs. Take, for example, the 2003 Northeastern United States blackout or the Texas power crisis of 2021, triggered by an unprecedented winter storm. Beyond leaving millions without electricity, these calamities caused severe food and water shortages, exacerbated heating challenges during frigid conditions, and tragically led to the loss of over two hundred lives. Another illustrative incident is the 2011 Southwest Blackout. Originating from a singular transmission line malfunction, it rapidly magnified into a cascading grid failure, inflicting an economic toll approximated at 100 million dollars. They serve as stark reminders of the imperative nature of power grids as safety-critical systems demanding stringent performance guarantees.

The need to avoid such catastrophic failures and ensure resilient power delivery has led to a deepened interest in the study of state estimation. This indispensable tool provides a comprehensive observability of power systems, enabling effective decision-making processes that enhance grid operations. State estimation employs advanced algorithms, computational techniques, and statistical models to determine the underlying states of a power system. It consolidates varied measurements

and reconciles discrepancies, overcoming the challenges of imperfect or incomplete data to present an accurate view of the system's status. As a result, real-time situational awareness from state estimation gives power system operators the foresight they need to spot, predict, and fix possible anomalies, thereby mitigating the damage they cause. The growing inter-connectivity of power networks, incorporation of renewable energy sources, the proliferation of distributed generation, and other technological advancements make it even more critical to ensure the system is stable, competent, and secure.

In power system management, the SCADA (Supervisory Control And Data Acquisition) system plays a pivotal role in data acquisition and real-time monitoring. As shown in Fig. 1, integral to SCADA's functionality are Remote Terminal Units (RTUs), devices that transform analog signals into digital data streams that become integrated into state estimation algorithms. There is also a growing number of Phasor Measurement Units (PMUs), supported by the Global Positioning System (GPS) for precise time-stamping, which provide high-resolution, time-synchronized measurements of voltage and current phasors. PSSE aims to recover the system's current underlying voltage phasors, given SCADA measurements, PMU measurements, and a system model based on assumed parameters [1]. State estimation not only helps prevent

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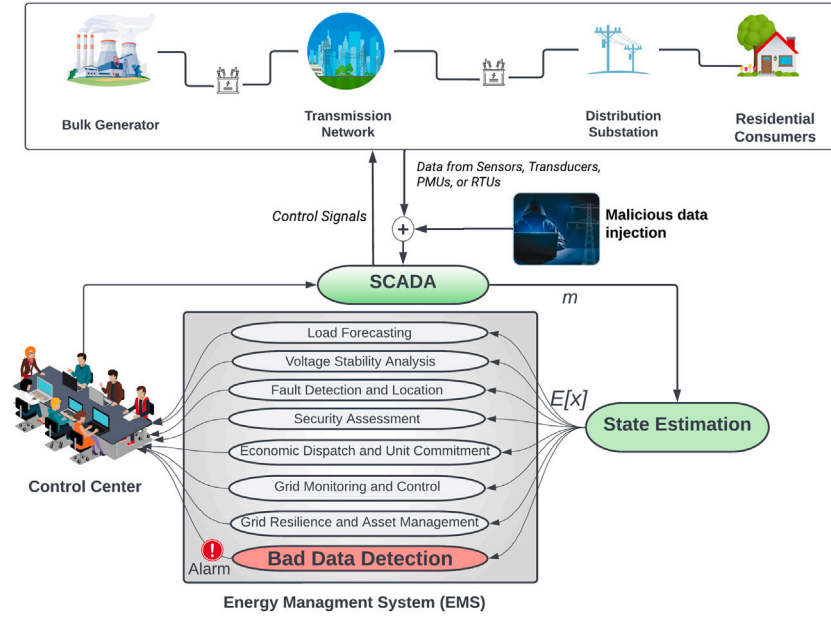


Fig. 1. An illustration of the operations of SCADA/EMS for power systems monitoring and control.

failures in the power network, but also underpins every aspect of real-time power system operation and control. The major challenges of state estimation arise from the fact that power system measurements are riddled with noise and corrupt data, and subject to missing values or cyber-attacks. Furthermore, due to the laws of physics, the measurement values are nonconvex functions of voltages, which limit the application of traditional convex optimization techniques that enjoy strong theoretical and performance guarantees.

The field of PSSE has seen the development and utilization of numerous algorithms and methodologies, specifically designed to tackle the various challenges and complexities inherent in the problem [2, 3]. Linear state estimation approaches are computationally efficient solutions that rely on the assumption of a linearized power system model [4,5]. Nevertheless, the aforementioned advantages are accompanied by a trade-off in terms of reduced accuracy in certain scenarios [6,7]. Conversely, nonlinear state estimation techniques, which consider the complete nonlinearity of the power flow equations, typically exhibit greater precision [8]. This precision comes, however, at the expense of increased computational demands [3,9]. The Weighted Least Squares (WLS) technique is a widely used technique for state estimation, which aims to minimize the sum of squared residuals between the measured values and the estimated values. This minimization is achieved by assigning different weights to the residuals, which are inversely proportional to the measurement error covariances [2]. The Weighted Least Squares (WLS) method achieves a desirable balance between computational efficiency and estimation accuracy, which has led to its widespread adoption in PSSE [5]. Another well-known technique is the Kalman filter, an iterative algorithm proficient in real-time state estimation amid noisy data and uncertainties [10,11]. The Kalman filter dynamically refines the state estimates through a prediction–correction mechanism [12]. To address more complex, non-linear system behaviors and uncertainties, various extensions and adaptations of this technique have been developed, such as the Unscented Kalman Filter for cases with unknown noise statistics [13], the Cubature Kalman Filter suitable for smart grids with communication constraints [14], and Extended Kalman Filter for handling non-linearities and spoofing attacks [15]. Other variants, such as Particle Filter and Gaussian Mixture Filter, offer solutions for limited bandwidth and secure state estimation, respectively [16,17]. For a comprehensive review of classic methods for power system state estimation, we refer the reader to [18].

The advent of deep learning algorithms in recent years has signaled a new epoch in the field of PSSE, offering notable solutions to the electrical grid's complexities [19,20]. These advancements illustrate deep learning's capability to tackle complex and computationally efficient algorithms in power systems [21]. The work by [22] draws attention to a transformative change of trend in state estimation methodologies, transitioning from purely model-driven frameworks to hybrid-driven approaches. These data-driven methodologies, distinguished by their ability to automatically learn intricate patterns and relationships from vast historical and real-time data repositories, have emerged as powerful alternatives to traditional state estimation techniques. Meanwhile, [23] authors achieve the initialization of network voltages for non-linear least squares SE solver through the training of feed-forward neural networks. Graph-based approaches are ideally positioned to encapsulate the spatial dependencies and long-range interactions that underpin the functioning of the electrical grid [24]. They represent power networks as graphs, with buses constituting nodes and transmission lines acting as edges. These methods successfully exploit the interconnected nature of power systems to reveal their underlying dynamics using sophisticated architectures such as graph neural networks (GNNs) [25,26] and graph neural ordinary differential equations (GDEs) [27]. GNNs have had a significant impact on the field of power system analysis and modeling. These techniques, for example, have been used in the field of fault detection [28,29], device misconfigurations [30], parameters identification using variance uncertainty for the unbalanced branch loss function [31], distinguishing between distinct types of instabilities [32], and even detecting malicious data injection attacks [33]. GNNs have also demonstrated their transformative capabilities in the prediction and estimation of power flows, such as power flow analyses [34] and probabilistic power flow studies [35], suggesting that a trained GNN could effectively perform similar calculations with less computational overhead. Significantly, [36,37] have put forth methodologies based on GNNs with the objective of enhancing the precision and effectiveness of power flow predictions. On the other hand, additional studies conducted by [38] have expanded upon these methodologies, presenting more comprehensive techniques for power flow analysis via typed graph neural networks. In addition, several artificial neural networks (ANN) versions, including Recurrent Neural Network (RNN) [39], Long short-term memory (LSTM) [40], and Residual Neural Network (ResNet) [41] have been used to solve data-driven forecasting-assisted state estimation within smart grids. These

advanced algorithms often handle measurement data as multivariate time series. For instance, [39] authors addressed the state estimation challenge using RNNs by leveraging measurement data and historical voltage patterns. The research conducted in [42] shows that the authors integrated GNNs with conventional power system state estimation techniques, incorporating data-driven priors to improve the algorithm's resilience. Meanwhile, [20] authors emphasized the efficacy of temporal graph convolution networks for state estimation. They used a GNN with the gated recurrent unit to predict state variables, which was trained on a time series of node voltage measurements gathered from PMUs. Adding to this, [43] authors proposed the utilization of temporal graph convolution networks to highlight the dynamic characteristics of power systems and capture the temporal variations in node voltage measurements. In their notable studies [44,45], the authors proposed a GNN-based approach and a graph attention network (GAT) approach, respectively, implemented over a factor graph, offering robust and computationally efficient solutions adaptable to unstable scenarios and malicious data. These models condense the essence of grid topology, enhancing the state estimation process by taking into account the network's and its components' unique features.

In this work, we explore the potential of using the attention component from transformer models for PSSE. The attention module, introduced by [46], is the foundational element of the transformer model. It was proposed to address the limitations of recurrent and convolutional neural network models in handling sequential data, especially for Natural Language Processing (NLP) tasks. The attention component dynamically assigns importance to different elements in a sequence, allowing the model to focus selectively on relevant parts of the input data for a given task. This component has been instrumental in the development of various models beyond NLP, expanding its relevance to different fields due to its adeptness at identifying and underlining complex dependencies in various datasets. Our model, Graph Attention Estimation Network (GAEN), integrates the transformative attention component with Graph Convolutional Neural Networks (GCNNs) to serve as a data-driven nonlinear state estimator. Building upon the initial explorations in [23,26], we address their limitations by augmenting GCNNs with an additional set of layers of attention. The GAEN model exhibits computational efficiency, enabling it to scale effectively to large power networks. By employing supervised learning, GAEN demonstrates promising generalization capabilities to previously unseen measurement scenarios. The primary contributions of this paper can be summarized as follows:

- We employ graph attention to develop power system state estimators from SCADA and PMU measurements, offering an efficient model to tackle the complexities of state estimation.
- The incorporation of the attention mechanism captures the complex and non-linear relationships within the grid, resulting in enhanced feature representation and improved accuracy of state estimations.
- The adaptive nature of the GAEN model allows for handling ill-conditioned scenarios, and evolving dynamics in power systems, resulting in more robust state estimators.
- The selective focus on relevant features enables our architecture to scale efficiently with large power systems, addressing the computational challenges associated with state estimation in complex networks.
- Through numerical experiments, we demonstrate the superior performance of state estimators derived from GAEN compared to existing supervised learning techniques.

## 2. State estimation in power grids

As discussed in Section 1, state estimation is a fundamental problem in power systems, whose goal is to identify the underlying system voltages based on measurements across the grid. This continuous oversight

of the power network is vital to prevent adversities like blackouts, ensuring timely interventions when required. The current section offers a succinct overview of state estimation in power grids.

We represent the electrical network as a directed graph  $\mathcal{G} = \{\mathcal{B}, \mathcal{L}\}$ , where  $\mathcal{B} = \{b_1, \dots, b_N\}$  corresponds to the set of  $N$  buses and  $\mathcal{L} \subseteq \mathcal{B} \times \mathcal{B}$  denotes the interconnecting transmission lines. Although power can flow in either direction of each transmission line, the graph model assumes a predetermined reference direction. The neighborhood set of bus  $b_i$  is denoted as  $\mathcal{N}_i = \{b_j \in \mathcal{B} : (b_j, b_i) \in \mathcal{L} \text{ or } (b_i, b_j) \in \mathcal{L}\}$ . The incidence matrix,  $\mathbf{B} \in \mathbb{R}^{N \times M}$ , where  $M$  corresponds to the number of transmission lines, characterizes the network's connectivity by taking on a value of  $-1$  for from-nodes,  $+1$  for to-nodes, and  $0$  for the remaining elements. The complex-valued voltage at a network bus  $b_i$  is denoted by  $v_i \in \mathbb{C}$ , whose polar representation consists of its magnitude  $|v_i| \in \mathbb{R}^+$  and angle  $\angle v_i \in [-\pi, \pi]$ . The complex power injection at bus  $b_i$  is represented by  $s_i \in \mathbb{C}$ , with active power  $p_i \in \mathbb{R}$  and reactive power  $q_i \in \mathbb{R}$ , such that  $s_i = p_i + jq_i$ , with  $j$  as the imaginary unit.

The grid's physical characteristics are encapsulated in the admittance matrix. More specifically, each bus  $b_i$  is accompanied by a nodal shunt admittance  $\hat{y}_i = y_i^{\Re} + jy_i^{\Im}$ , while each line  $(b_i, b_j)$  is represented by a  $\Pi$ -model with a series admittance  $\tilde{y}_{ij} = \tilde{y}_{ij}^{\Re} + j\tilde{y}_{ij}^{\Im}$  and the total shunt admittance  $\check{y}_{ij} = \check{y}_{ij}^{\Re} + j\check{y}_{ij}^{\Im}$ . These values can be conveniently captured in the admittance matrix  $\mathbf{Y} \in \mathbb{C}^{N \times N}$  defined as

$$[\mathbf{Y}]_{ij} = \begin{cases} \hat{y}_i + \sum_{k: b_k \in \mathcal{N}_i} (\tilde{y}_{ik} + \check{y}_{ik}/2) & \text{if } i = j, \\ -\tilde{y}_{ij} & \text{if } (i, j) \in \mathcal{L} \text{ or } (j, i) \in \mathcal{L}, \\ 0 & \text{otherwise.} \end{cases}$$

Note that the admittance matrix can be decomposed into its real and imaginary parts  $\mathbf{Y} = \mathbf{Y}^{\Re} + j\mathbf{Y}^{\Im}$  for  $\mathbf{Y}^{\Re} \in \mathbb{R}^{N \times N}$  and  $\mathbf{Y}^{\Im} \in \mathbb{R}^{N \times N}$ . Using the admittance values, we can write the active and reactive powers flowing over line  $(i, j)$  as nonlinear functions of voltages in polar form:

$$p_{ij} = -y_{ij}^{\Re}(|v_i|^2 - |v_i||v_j|\cos(\angle v_{ij})) + y_{ij}^{\Im}|v_i||v_j|\sin(\angle v_{ij}) \quad (1a)$$

$$q_{ij} = y_{ij}^{\Im}(|v_i|^2 - |v_i||v_j|\cos(\angle v_{ij})) + y_{ij}^{\Re}|v_i||v_j|\sin(\angle v_{ij}) \quad (1b)$$

where  $y_{ij}^{\Re} = [\mathbf{Y}^{\Re}]_{ij}$ ,  $y_{ij}^{\Im} = [\mathbf{Y}^{\Im}]_{ij}$ . The term  $\angle v_{ij}$  denotes the phase angle difference between the voltages at buses  $i$  and  $j$ . This phase angle difference, expressed as  $\angle v_i - \angle v_j$ , is important in determining the active and reactive power flow between buses. Similarly, the active and reactive nodal power injections at bus  $b_i$  can be written as nonlinear functions of voltages in polar form:

$$p_i = \sum_{j=1}^N |v_i||v_j| (y_{ij}^{\Re} \cos \angle v_{ij} + y_{ij}^{\Im} \sin \angle v_{ij}) \quad (2a)$$

$$q_i = \sum_{j=1}^N |v_i||v_j| (y_{ij}^{\Im} \sin \angle v_{ij} - y_{ij}^{\Re} \cos \angle v_{ij}) \quad (2b)$$

The summations effectively only include buses sharing a line with bus  $b_i$  and itself due to the admittance matrix's sparsity.

The SCADA system receives measurements from RTU devices around the grid. The measurement types cover nodal active and reactive power injections, branch active and reactive power flows, and voltage magnitudes. PMU devices can measure voltage phasors directly (i.e., voltage magnitude and angle). In practice, the obtained measurement values do not exactly match the power flow equations in (1) and (2) due to the presence of noise and bad data. In other words, the relationship between the measurements and the state variables can be expressed as  $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{e} + \mathbf{a}$ , where  $\mathbf{z}$  denotes the vector of all measurement values,  $\mathbf{x}$  denotes the vector of voltages,  $\mathbf{h}(\mathbf{x})$  denotes the vector of measurement functions,  $\mathbf{e}$  represents random measurement noise, and  $\mathbf{a}$  corresponds to the sparse gross errors. Then, the objective of state estimation is to determine the true state  $\mathbf{x}$  from the set of noisy and potentially incomplete measurements. As we mentioned earlier, various techniques have been proposed to address this problem, including weighted least

squares (WLS), maximum likelihood estimation (MLE), and robust state estimation methods. These techniques aim to minimize the impact of measurement noise and gross errors while considering the nonlinear relationship between the measurements and the state variables.

Utilizing measurements of power injections, line flows, and other relevant data, we adopt graph-based methodologies that consider both the system's inherent physical attributes and its topological characteristics. In our model, line power flow measurements are specifically treated as line features. Knowing that power flow can be observed from both sides of a line, our model incorporates four distinct types of line measurements, enhancing the accuracy and robustness of our analysis.

### 3. Graph Attention Estimation Network (GAEN) model

Numerous factors contribute to the complexity of state estimation. Among the most salient are the inherent nonlinearity of power flow equations, measurement noise, and the presence of bad data. An efficacious state estimation algorithm seeks to capitalize on the network's connectivity information and the redundancy of measurement data to deduce the underlying state that best aligns with the measurement values, despite these challenges. Let us consider a graph  $\mathcal{G} = \{\mathcal{B}, \mathcal{L}\}$ , where a graph signal possessing  $F$  features can be denoted by a matrix  $\mathbf{X} \in \mathbb{R}^{N \times F}$ , with the  $i$ th row corresponding to the feature values at the  $i$ th node  $\mathbf{x}_i \in \mathbb{R}^F$ . Although the matrix representation of graph signals is mathematically convenient, it fails to keep track of the information pertaining to the underlying graph structure. To address this shortcoming, we use the graph matrix description (GMD)  $\mathbf{S} \in \mathbb{R}^{N \times N}$ , which shares the sparsity pattern of the underlying graph and utilizes common choices such as Laplacian, Markov, and adjacency matrices [47].

The GMD facilitates the definition of a fundamental linear operation between graph signals,  $\mathcal{S} : \mathbb{R}^{N \times F} \rightarrow \mathbb{R}^{N \times G}$  [47]. Given the filter coefficients,  $\mathbf{H}_k \in \mathbb{R}^{F \times G}$ , the graph filters such as the finite impulse response (FIR),  $(\sum_{k=0}^K \mathbf{S}^k \mathbf{X} \mathbf{H}_k)$ , are constructed based on this operation which enables a distributed computation where each node calculates an output based on information exchanged with their  $k$ -hop neighbors. Nevertheless, such graph filters are linear and restricted to capturing linear relationships between input and output. Nonlinear relationships can be embedded using the GCNNs  $\Phi : \mathbb{R}^{N \times F} \rightarrow \mathbb{R}^{N \times G}$ , defined as a cascade of  $L$  layers that each apply a graph filter followed by a nonlinear and pointwise activation function. The GCNNs exhibit favorable properties such as permutation invariance, Lipschitz continuity concerning changes in the GMD  $\mathbf{S}$ , and improved processing capabilities compared to linear graph filters. A more detailed discussion on the GCNN architecture can be found in Section 3 of [26].

Our GAEN model comprises multiple layers designed to process the nodal, line, and PMU measurements. All layers, before applying the attention mechanism layer, apply a graph filter and a corresponding nonlinear activation function. The difference between layers lies in the specific filter and activation function used. The model's final layer produces the estimated state, which is then compared to the true state for performance evaluation. A loss function, such as the mean squared error (MSE), is minimized to learn the filter coefficients using a training dataset that includes the nodal, line, and PMU measurements along with the corresponding true states. The design choices of the GAEN model, including the number of layers  $L$ , the order  $K$  of filters, the number of features  $F_\ell$  at each layer, the pointwise activation function  $\sigma$ , etc. serving as hyperparameters, are selected based on cross-validation, model selection techniques, and tuning.

As shown in Fig. 2, our designed model consists of three blocks of layers with the objective of processing the input nodal and line measurement features. In the following subsections, we provide more details of our multi-layered structure, focusing on refining and leveraging both node and edge (line) features. Section 3.1 concentrates on processing node features using a series of graph convolution operations. On the other hand, in edge filtering layers (sub Section 3.2), we apply similar methodologies but specifically tailored for edge features,

and with an extra processing to transform edge signals back to node signals. Once the initial processing through filtering layers is complete, in sub Section 3.3, we provide details of the multi-head attention mechanism. This mechanism transforms and further refines the node and edge features into query, key, and value matrices. Through a series of transformations, scaling, and attention computations, we generate a combined, weighted output. These operations ensure that the model focuses on significant features, making the estimation process more accurate. Finally, we process the aggregated attention-weighted features in the readout layer to generate the voltage estimates, encapsulating both nodal and line measurements. The incorporation of PMU measurements into the training process is facilitated by adding penalty terms to the loss function and is explained in sub Section 4.1.

#### 3.1. Node filtering block

The first block is designed to process nodal measurement signals. The first layer of this block applies a graph filter with a predetermined order  $K$ , followed by a nonlinear and pointwise activation function  $\sigma$ . The output of the first layer is passed through the subsequent layers with similar operations, using the filter order  $K$  and activation functions  $\sigma$  accordingly. The node filtering layers operate on the node features of the input graph using a series of graph convolution operations defined as follows:

$$\mathbf{X}_\ell = \sigma \left( \sum_{e=1}^2 \sum_{k=0}^K \mathbf{S}_e^k \mathbf{X}_{\ell-1} \mathbf{H}_{\ell k}^e \right) \quad \ell = 1, 2, \dots, L \quad (3)$$

where  $\mathbf{X}_\ell \in \mathbb{R}^{N \times F_\ell}$  denotes the node graph signals with  $F_\ell$  features, and  $\mathbf{H}_{\ell k} \in \mathbb{R}^{F_{\ell-1} \times F_\ell}$  are the filter coefficients. The values  $k$  and  $e$  refer to the GMD matrix power and the additional index accounting for multiple GMDs on the filter matrix, respectively. The output of this layer offers a filtered representation of the node features. The  $\{\mathbf{H}_{\ell k} \mid k = 0, \dots, K\}$  set contains the  $K+1$  filter coefficients, where combined with the  $\mathbf{S}$  matrix represents the graph filter. The graph filter's order,  $K$ , dictates the depth of information exchange in the graph or, in other words, the maximum number of neighboring hops a node communicates with. For an order  $K = 1$ , a node gathers information only from its immediate neighbors. When  $K = 2$ , it factors in data from both immediate neighbors and their adjacent nodes. This ensures that even without complete knowledge of the full matrix  $\mathbf{S}$ , nodes can effectively exchange information locally and capture expansive contextual insights from their surrounding nodes by simply varying  $K$ .

**Remark 1.** The choice of the GMD plays an important role in highlighting specific attributes of the underlying graph. While widely used matrices such as the Laplacian, Markov, and adjacency matrices are able to encode the graph's connectivity, they cannot capture the admittance values (which can be viewed as the strengths of the connections) of the power grid. In this paper, instead of having a single GMD for the node filtering block, we propose using two GMDs:  $\mathbf{S}_1 = \Re(\mathbf{Y})$ ,  $\mathbf{S}_2 = \Im(\mathbf{Y})$ . By doing this, we account for two processes, one for active (real) power and another for reactive (imaginary) power.

#### 3.2. Edge filtering block

The second block consists of edge filtering layers and performs a similar role as the previous block but operates on the edge features instead of the nodal features. In these layers, graph convolutions are used to process the edge features as follows:

$$\bar{\mathbf{X}}_\ell = \sigma \left( \sum_{k=0}^K \bar{\mathbf{S}}^k \bar{\mathbf{X}}_{\ell-1} \bar{\mathbf{H}}_{\ell k} \right) \quad \ell = 1, 2, \dots, L \quad (4)$$

where  $\bar{\mathbf{X}}_\ell \in \mathbb{R}^{M \times \bar{F}_\ell}$  denotes the edge graph signals with  $\bar{F}_\ell$  features,  $\bar{\mathbf{S}}^k \in \mathbb{R}^{M \times M}$  represents the GMD matrix for line interactions in a graph, adopting the graph's sparsity pattern, such that  $\bar{\mathbf{S}}^k$ 's elements

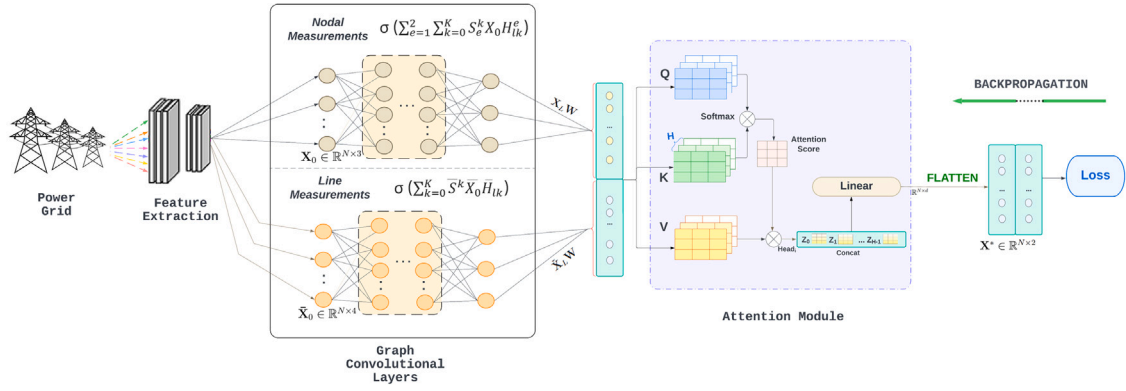


Fig. 2. Graph Attention Estimation Network Architecture.

are zero when the corresponding graph lines do not intersect, and  $\tilde{\mathbf{H}}_{\ell k} \in \mathbb{R}^{F_{\ell-1} \times F_{\ell}}$  are the filter coefficients. Finally, we convert the edge signals into node signals by multiplying the output edge graph signals of the last layer  $\tilde{\mathbf{X}}_L$  by the incidence matrix  $\mathbf{B}$ :

$$\tilde{\mathbf{X}}_L = \mathbf{B}\tilde{\mathbf{X}}_L \quad (5)$$

Note that we use three layers ( $L = 3$ ) for both the node filtering block and the edge filter block. However, this is purely a design choice and can be altered by the user's discretion.

**Remark 2.** As opposed to using two GMDs for the node filtering block, we use a single GMD for the edge filtering block. We choose between two different options: the linegraph Laplacian matrix or the Hodge Laplacian matrix. The *linegraph Laplacian* [48] GMD matrix is formed by converting edges from the original graph into nodes and the intersections between these edges into new edges, thereby effectively capturing the incidence between edges. On the other hand, the Hodge Laplacian [49,50], offers a unique view, defined in terms of the graph's incidence matrix  $\mathbf{B}$ , linking the dual nature of nodes and edges. Both choices preserve the same sparsity patterns and offer a clearer understanding of network connectivity and interactions.

### 3.3. Multi-head attention block

In the third block, the features obtained from the node and edge filtering blocks are processed by a multi-head attention mechanism. The Multihead Attention module starts by transforming the node signals  $\mathbf{X}_L$  and edge signals (converted into nodal information via eqn. (5))  $\tilde{\mathbf{X}}_L$  into Query ( $\mathbf{Q}$ ), Key ( $\mathbf{K}$ ), and Value ( $\mathbf{V}$ ) representations through three distinct learnable linear transformations. We transform both  $\mathbf{Q}$  and  $\mathbf{K}$  from the node signals  $\mathbf{X}_L$ , as it enables the model to compare different node signals, enabling the attention component to distinguish and weight the importance of different nodes in the context of others. This helps in capturing the underlying structure and extracting relevant features from the nodal signals. Meanwhile,  $\mathbf{V}$  is mapped to the output transformation of edge graph signals of the last layer  $\tilde{\mathbf{X}}_L$  with the incidence matrix  $\mathbf{B}$ , allowing the model to incorporate contextual information effectively. This differentiation in mapping is important for balancing the contribution of individual node signals and the contextual information from their connections  $\tilde{\mathbf{X}}_L$ , providing a more in-depth representation. These attention heads operate independently, as parallel projection spaces to capture different aspects in each head, and their outputs are concatenated and linearly transformed to yield the final output of the attention mechanism. These transformations are as follows:

$$\mathbf{Q} = \mathbf{X}_L \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}_L \mathbf{W}_K, \quad \mathbf{V} = \tilde{\mathbf{X}}_L \mathbf{W}_V \quad (6)$$

where  $\mathbf{W}_Q$ ,  $\mathbf{W}_K$ , and  $\mathbf{W}_V \in \mathbb{R}^{F_L \times F_L}$  are the learnable weight matrices, and  $F_L$  is the feature dimension of the last layer of the node filtering

and edge filtering blocks. Each of these transformed matrices is further processed and the scaled dot-product attention for each head is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \mathbf{V} \quad (7)$$

where  $d$  represents a uniform dimension for the keys across all heads, which is set to the feature dimension divided by the number of heads ( $F/H$ ). Subsequently, the matrices  $\mathbf{Q}$ ,  $\mathbf{K}$ , and  $\mathbf{V}$  are reshaped from having dimensions  $(N, F_L)$  to  $(N, H, F_L/H)$  by splitting them into multiple heads, where  $H$  denotes the number of heads. The softmax function is applied to ensure that the output of the attention mechanism is a proper probability distribution. After computing the attention output for each head, where each head operates as follows:

$$\mathbf{Z}_i = \text{Attention}(\mathbf{Q}\hat{\mathbf{W}}_{Q_i}, \mathbf{K}\hat{\mathbf{W}}_{K_i}, \mathbf{V}\hat{\mathbf{W}}_{V_i}) \quad (8)$$

then the outputs are concatenated together and linearly transformed to yield the final output:

$$\mathbf{O} = \text{Concat}(\mathbf{Z}_1, \dots, \mathbf{Z}_H) \mathbf{W}_O \quad (9)$$

where  $\mathbf{W}_O$  is another learned weight matrix. After the attention mechanism, we head towards the readout layer by reshaping the output of the attention mechanism. We apply a linear layer and a softmax function to generate the final output of the model. The readout layer is designed to aggregate the attention-weighted features, derived from both nodal and line measurements into the final voltage estimates.

## 4. Numerical results and discussion

### 4.1. Model training

In this section, the training process of the GAEN model is outlined, and the trained model is subsequently tested on various examples to assess its accuracy and robustness under measurement data loss, which may result from absence of measurement sensors or communication failures, and under cyber-attacks in the form of malicious data injections.

We generated datasets from IEEE power system test cases, each containing samples of voltage magnitudes and angles as true states. The magnitudes are sampled from a normal distribution with a mean of 1 and a standard deviation of 0.02, while angles are sampled from a normal distribution with a mean of 0 and a standard deviation of 10. These values, used as inputs to power flow equations and added with Gaussian noise, form labeled datasets. Of these, 81% is allocated for training, 9% for validation, and the remaining 10% for testing. The training process involves minimizing the MSE loss function between the true states and the estimated states produced by each of the models. In order to incorporate information obtained from PMU measurements,

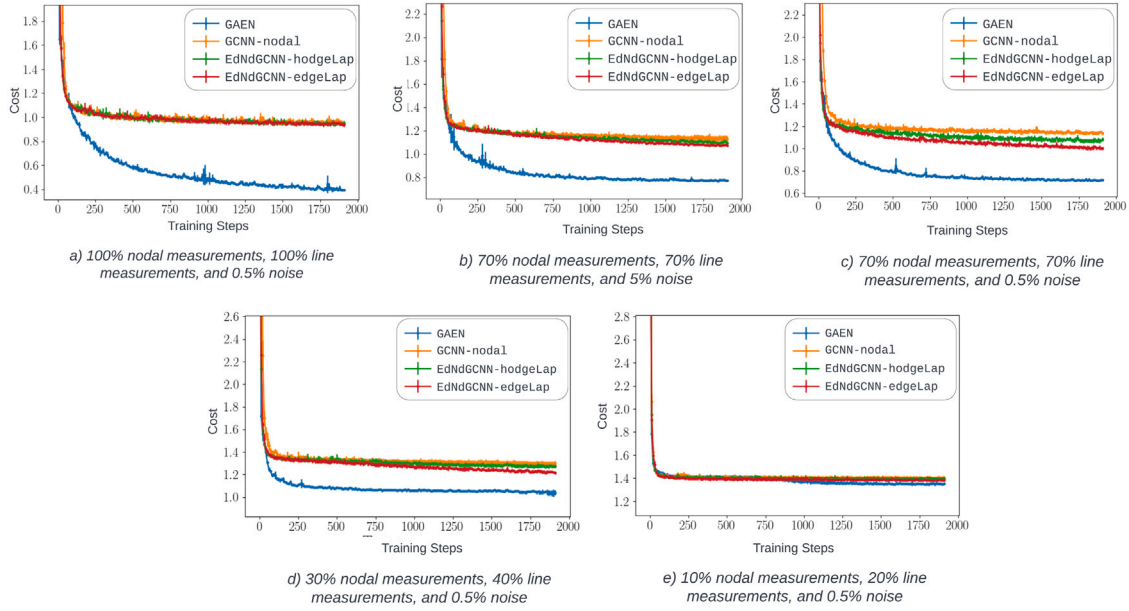


Fig. 3. Cost evaluation during training for GCNN-nodal, EdNdGCNN-hodgeLap, EdNdGCNN-edgeLap, and GAEN model under varying nodal and line measurement percentages with different noise levels. (a) Complete nodal and line measurements with minimal noise; (b) Moderate measurements with increased noise; (c) Moderate measurements with minimal noise; (d) Reduced measurements with minimal noise; (e) Minimal measurements with minimal noise.

for each node with a PMU measurement, the deviation of the estimated state from the PMU measurement can be added as a penalty in the loss function. The performance of a trained GAEN model is assessed based on its capacity to solve the state estimation task on unobserved data. As we can see in Fig. 3, validation results under diverse conditions – from full nodal and line measurements with minimal noise to reduced measurements with varying noise levels – reveal that the GAEN model estimators not only learn effectively across training steps but also outperform other existing GCNN models. Also, the trained architecture is then validated on a separate dataset to assess its performance in the presence of noise and potential bad data (cyber-attack), as detailed in Section 4.4.

As for the specific training parameters, we used a batch size of 128, the ADAM optimizer, enabled learning rate decay with a decay period of five epochs and a decay rate of 0.95, four data splits, and 40 epochs. For the dimensions of our GAEN model, the following parameters were selected: node signal dimensions of [3, 60, 240, 312] (this includes the number of features for the input signal and the outputs of the three hidden layers), line signal dimensions of [4, 80, 320, 312], number of attention heads [5, 5, 5] on each layer, the hyperbolic tangent function (Tanh) as the nonlinear activation function, and no dropout.

#### 4.2. Comparison against existing methods

To evaluate the performance of the GAEN model, we conducted numerical simulations on the IEEE 39-bus system. We compared its efficacy against three GCNN variants: (i) *GCNN-nodal*, which relies solely on nodal measurements; (ii) *EdNdGCNN-hodgeLap*, which incorporates both nodal and line measurements using Hodge Laplacian  $\tilde{S} = B^T B$  as its edge GMD, which means that  $\tilde{S}$  is described in terms of the graph incidence matrix  $B$ ; and (iii) *EdNdGCNN-edgeLap*, which, while also considering both nodal and line measurements, uses the *linegraph Laplacian* as its edge GMD. Once trained, the performance of the GAEN, GCNN-nodal, EdNdGCNN-hodgeLap, and EdNdGCNN-edgeLap models is assessed on the test dataset to analyze their robustness to noise and cyber-attacks.

#### 4.3. Numerical simulations and analysis

In this subsection, the performance evaluation of the Graph Attention-based Estimation Network (GAEN) model is presented in terms of Root Mean Squared Error (RMSE) values under varying percentages of line and nodal measurements, with a noise level of 0.5%. The RMSE values were averaged over four data splits and computed for the IEEE 39-bus test case, with measurement percentages spanning from 100% to 30% for line measurements and 100% to 70% for nodal measurements.

For the scenario with 100% nodal measurements, the GAEN model exhibited the lowest RMSE values across all line measurement percentages, thus signifying superior performance in state estimation accuracy. The RMSE values for the GAEN model were observed within the range of  $8.59 \times 10^{-3}$  to  $8.71 \times 10^{-3}$ . As shown in Fig. 4(a), in contrast, the other algorithms GCNN-nodal, EdNdGCNN-hodgeLap, and EdNdGCNN-edgeLap, demonstrated higher RMSE values, indicating a reduced level of accuracy compared to the GAEN model. Upon reducing the nodal measurements percentage to 90% (Fig. 4(b)), the GAEN model continued to maintain its lead in performance, with RMSE values ranging from  $1.083 \times 10^{-2}$  to  $1.48 \times 10^{-2}$ . The other algorithms persisted in exhibiting higher RMSE values, thereby reinforcing the superior performance of the GAEN model. With a further reduction in nodal measurements to 80%, as shown in Fig. 4(c), the RMSE values for the GAEN model increased, varying between  $1.286 \times 10^{-2}$  and  $2.111 \times 10^{-2}$ . Nevertheless, it still outperformed the other algorithms in terms of accuracy. Finally, for the case with 70% nodal measurements, the GAEN model's RMSE values fluctuated between  $1.512 \times 10^{-2}$  and  $1.963 \times 10^{-2}$ , which indicated a slight decrease in performance compared to the previous cases. However, the GAEN model remained the most accurate algorithm among the four considered.

The GAEN model was further assessed using a 5% noise level, revealing its performance under higher ambiguity in data. As we can see from Fig. 5, the GAEN model consistently exhibited the lowest RMSE values, indicating superior state estimation accuracy compared to the other algorithms. As the nodal measurements percentage decreased, the RMSE values for all algorithms increased, highlighting a decline in accuracy. However, the GAEN model maintained its lead in performance, showcasing its resilience and robustness. Similar trends were observed for line measurement percentage reductions. Despite this, the GAEN



Fig. 4. Performance evaluation of the GAEN model using RMSE values under varying percentages of line and nodal measurements, with a 0.5% noise level, for the IEEE 39-bus test case.

model retained its superior performance, demonstrating its adaptability to varying measurement conditions. Under a 5% noise level, the GAEN model displayed robust performance and accuracy in state estimation across different percentages of line and nodal measurements. The observed trends highlight the model's ability to remain stable and adapt to changing measurement conditions, further emphasizing its potential for state estimation applications. The GAEN model demonstrated superior performance in state estimation over other algorithms at both 0.5% and 5% noise levels, consistently achieving the lowest RMSE values. Despite the general trend of increased RMSEs at the higher 5% noise level across all algorithms, the GAEN model's performance remained superior, highlighting its robustness and adaptability to different noise environments and reinforcing its potential for effective state estimation in varied conditions.

In Fig. 6(a), we observe the performance dynamics under a scenario with limited nodal measurements (fixed at 30%) while varying line measurements from 100% to 30%, at a noise level of 0.5%. Compared to the previous results, where the nodal measurements percentage was higher (70% to 100%), the RMSE values for all algorithms increased, indicating a decrease in state estimation accuracy as fewer nodal measurements were available for the estimation process. Nevertheless, the GAEN model continued to outperform the other algorithms, maintaining the lowest RMSE values across all line measurement percentages. This demonstrates the GAEN model's ability to extract valuable information from edge signals, especially in scenarios with limited nodal data. This capability is crucial in environments with limited access to nodal information, highlighting its robustness and adaptability in state estimation applications under diverse and challenging measurement conditions.

As shown in Fig. 6(b), all these models have further been tested under an extreme adverse scenario with only 10% nodal measurements available, while the line measurements vary between 100% and 10%. From the results, it is evident that the GAEN model outperforms the other algorithms at higher percentages of line measurements. For instance, at 100% line measurements, the RMSE for the GAEN model

is 0.7258, which is significantly lower than the RMSE for GCNN-nodal (1.4036), EdNdGCNN-hodgeLap (1.1761), and EdNdGCNN-edgeLap (1.0886). This suggests that the GAEN model is more effective in utilizing all the line measurements to provide a more accurate state estimation. As the percentage of line measurements increases, the RMSE of the GAEN model decreases gradually, signifying that it is sensitive to the reduction in line measurements. This rise in RMSE is relatively slower until the line measurements reach 60%, after which there is a noticeable jump, particularly between 30% (RMSE: 0.946) and 20% (RMSE: 1.3277). This suggests that all the models may start to struggle when the line measurements fall below 30%, where their performance deteriorates rapidly. Comparing this performance to other models, GCNN-nodal shows relatively stable RMSE values across all percentages of line measurements but consistently performs worse than the GAEN model until line measurements fall below 30%. However, despite its stability, GCNN-nodal's RMSE is quite high, indicating less accurate state estimation. EdNdGCNN-hodgeLap and EdNdGCNN-edgeLap both start with higher RMSE values than GAEN model but display a steady increase in RMSE as the line measurements decrease. While the GAEN model excels at higher line measurement percentages, its performance gap with the other algorithms decreases as the information for line measurements gets limited.

#### 4.4. Performance analysis under malicious data injections

Cybersecurity has emerged as a critical issue in modern power systems. As the transition to smart grids generates an increasing abundance of data through sensors and electronic devices, the system's susceptibility to cyber-attacks intensifies. Although the wealth of information presents opportunities for informed decision-making, it is imperative to develop advanced data-analytic tools for detecting and mitigating such threats. The integration of additional components into the power grid further underscores the demand for computationally tractable methods, ultimately aiming to ensure the security, stability, and resilience of power systems against evolving cyber threats.

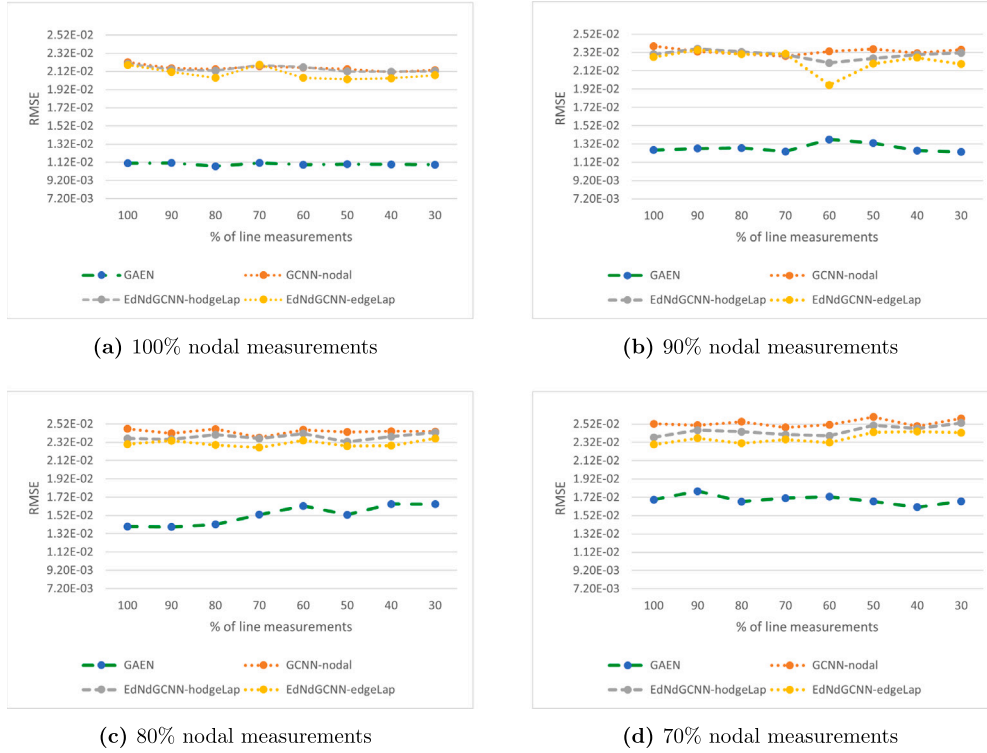


Fig. 5. Performance evaluation of the GAEN model using RMSE values under varying percentages of line and nodal measurements with a 5% noise level.

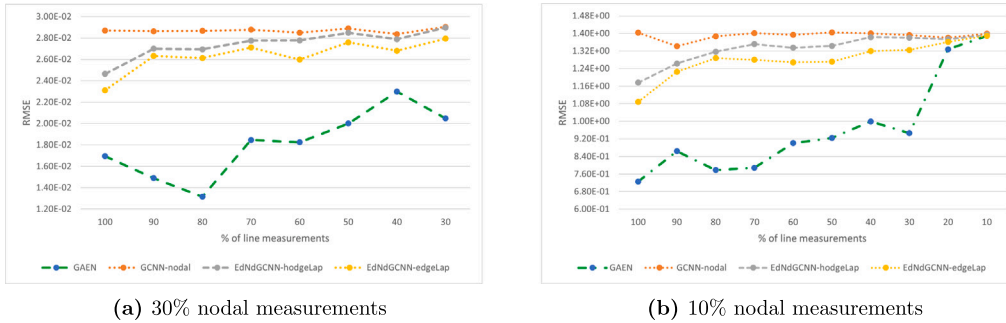


Fig. 6. This figure compares the GAEN model with GCNN-nodal, GCNN-hodgeLap, and GCNN-edgeLap models using a fixed nodal measurements percentage (30% or 10%) while varying line measurements percentages from 100% to 30% or from 100% to 10%, and the noise level was set at 0.5% on the IEEE 39 bus system.

This subsection presents a comparative performance evaluation of the GAEN model and the Nominal Weighted Least Squares (WLS) method under the context of cyber-attacks. The chosen benchmark for this comparative analysis is the IEEE 57-bus system. Each method's performance is evaluated in terms of the state estimation error that characterizes the discrepancy between the true state of the system and the estimated state, given measurements that might be subjected to data injection attacks. To simulate such an attack, measurement values at buses 2, 15, 16, and 17 are deliberately manipulated. Initially, the true states of the test samples, the measurements of the test samples, and the solution obtained from the GAEN method are loaded. The initial estimation error, before any optimization, is calculated for both the real and imaginary parts of the estimated state from the GAEN model. After preparing the measurement set, the system matrix, and the measurement vectors, a nonlinear least squares solver is used to perform optimization using the Levenberg–Marquardt algorithm [51]. This solver minimizes the least-squares differences between the estimated state variables and their corresponding true states.

To compare the robustness and performance of the GAEN method and the nominal WLS method, two initial points are considered for

each state estimation. One is derived from the GAEN method, while the other one is a nominal point typically used in WLS. These points are used as initial guesses for the least-squares solver to optimize the state estimation problem under the L2-norm. The state estimation error for each method is recorded and aggregated for further analysis. The square of the error is used as a metric of evaluation, considering both the real and imaginary parts of the complex power system state variables. For better visualization of the results, the logarithm (base 10) of the state estimation errors is plotted on the IEEE 57-bus system graph, where the nodes represent the buses and the edges represent the transmission lines. Our results revealed that the GAEN-based method consistently produced lower state estimation errors in the face of cyber-attacks compared to the nominal WLS method. This outcome, as shown in Fig. 7, demonstrated by the lower node weights in the graph for GAEN, indicates that the GAEN approach has a higher level of robustness against data injection attacks than the traditional WLS method. Blue represents a low error, while yellow signifies a high error. The localized impact of cyber attacks, as opposed to widespread effects observed with conventional optimization techniques, can be attributed to the distributed nature of computation inherent in neural networks.

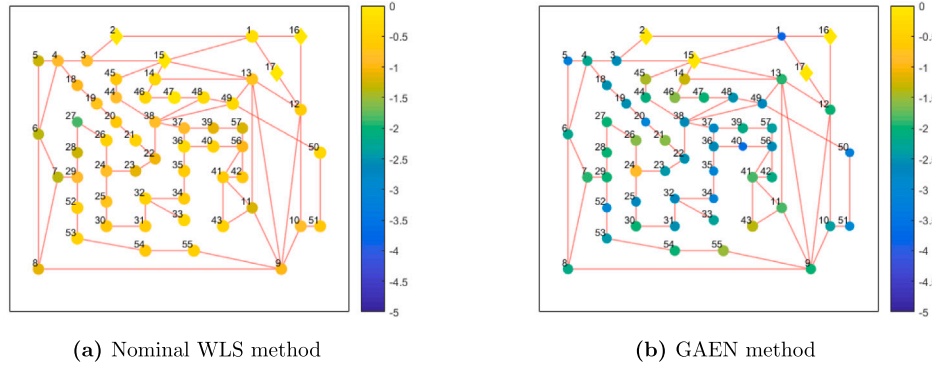


Fig. 7. Evaluation of the effectiveness of the Graph Attention Estimation Network (GAEN) and Weighted Least Squares (WLS) methods in estimating the state under malicious data injection.

This comparative analysis primarily demonstrates the GAEN model's effectiveness in minimizing state estimation errors under cyber-attacks, outperforming the traditional WLS method. This emphasizes the value of machine learning-based approaches like GAEN in enhancing the security and reliability of power systems against cyber threats.

## 5. Conclusions

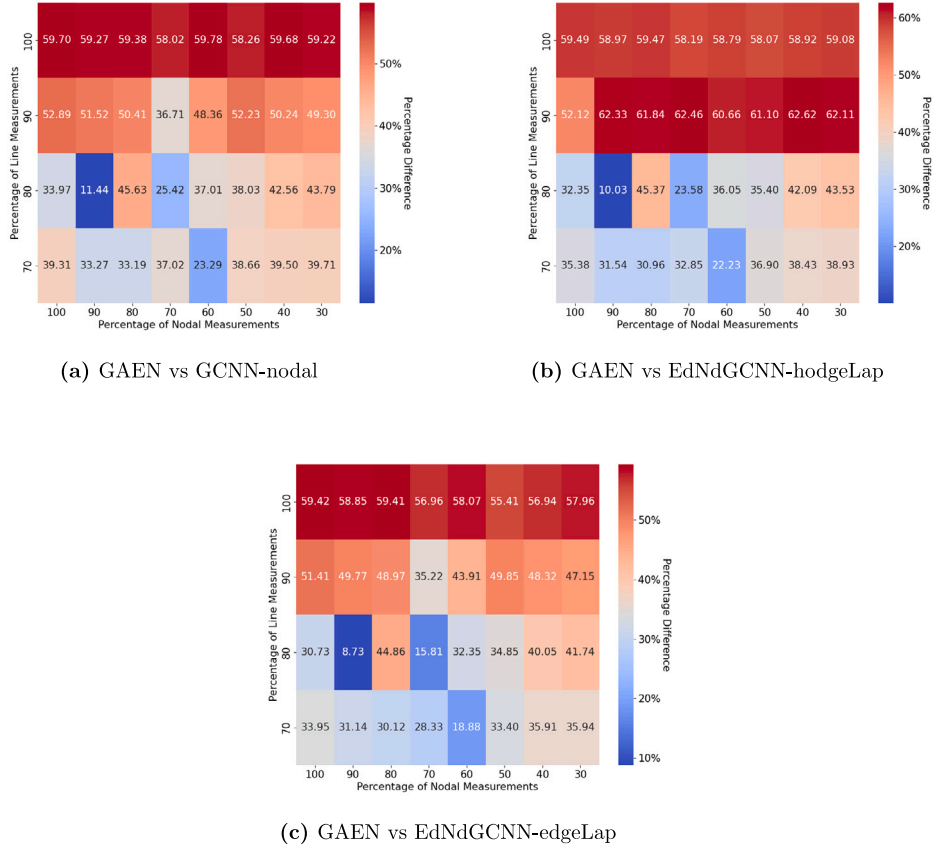
The GAEN model demonstrated its efficacy by focusing on relevant features and capturing intricate patterns within the data. The performance on validation sets was satisfactory, presenting a comprehensive evaluation of the GAEN model's performance in state estimation under various conditions, including different nodal and line measurement percentages and noise levels. The results consistently demonstrate the GAEN model's superior performance across the examined scenarios compared to other models, such as GCNN with only nodal measurements, GCNN with line measurements using the Hodge-Laplacian matrix, and GCNN with line measurements using line graph-Laplacian. Moreover, the GAEN model exhibits a robust performance even when the availability of nodal measurements is reduced, highlighting its adaptability in handling diverse measurement conditions. These findings show that the GAEN model is a promising architecture for state estimation in power systems, offering improved accuracy and resilience to varying network conditions. Future research may explore further refinements to the model and the application of the GAEN model in real-world power systems to validate its practical utility in dynamic environments. Tackling integration challenges is crucial in order to mitigate potential issues when incorporating the proposed model within existing power system infrastructure and technologies. Also, enhancing cyber-resilience warrants investigation, specifically by examining the proposed method's performance against a broader spectrum of cyber-attacks, encompassing more sophisticated and coordinated assaults. Lastly, the exploration of hybrid methods, combining the proposed approach with other machine learning techniques, such as reinforcement learning, further improves accuracy, efficiency, and resilience in PSSE, contributing to a more robust and reliable power grid.

## Acknowledgments

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## Appendix. Detailed results of the GAEN model against GCNN variants

In this appendix, we present a more detailed comparative evaluation of the GAEN model against several GCNN variants. The differences are precisely detailed in Table A.1 and Table A.2, which outline the averaged RMSE metrics under four data splits with line measurements ranging from 100% to 30% and nodal measurements from 100% to 70%, all subjected to a noise level of 0.5% and 5%, respectively. The percentage difference in RMSE of the GAEN model compared to the GCNNs variants for the IEEE 39-bus case using 0.5% noise level, illustrated in Fig. A.8(a), unveil the superior performance of the GAEN model, particularly in scenarios involving full line data, maintaining around a 59% improvement over the GCNN-nodal. This lead persists but lessens as nodal measurements decrease, with a significant drop to an 11.44% advantage at 90% nodal and 80% line measurements. At 70% line data, the advantage varies, dipping to 23.29% at 60% nodal measurements, indicating the GAEN model's robustness in sparse data scenarios. The performance gap between the GAEN and EdNdGCNN-hodgeLap Fig. A.8(b) tends to fluctuate as line measurements reduce from 100% to 70%. Initially, at 100% line measurements, the gap favors GAEN by a margin of 59.49% at full nodal capacity, indicating a strong lead. Meanwhile, at 90% line data, GAEN's advantage amplifies as node measurements decrease, peaking notably above 62% attributed to the EdNdGCNN-hodgeLap's oscillations in the averaged RMSE values. Moving to 80% line measurements, this lead slightly diminishes, yet GAEN still retains a notable advantage. Meanwhile for the GAEN vs EdNdGCNN-edgeLap models Fig. A.8(c), we see a slight decrease in GAEN's performance advantage as the percentage of line measurements declines from 100% to 70%. Starting from a strong lead at 100% line measurements, the performance advantage of the GAEN model demonstrates resilience but with a marginal decline through 90% and 80%. As we reach 70% line measurements, the GAEN model's edge over EdNdGCNN-edgeLap still shows considerable outperformance, indicating that the GAEN model retains its efficacy across a range of line data completeness.



**Fig. A.8.** Comparative analysis of the GAEN model against GCNN variants across varying nodal and line measurement scenarios in the IEEE 39-bus test case, using 0.5% noise level.

**Table A.1**

Evaluating GAEN model accuracy: Detailed averaged RMSE values over four data splits across 100% to 30% line measurements and 100% to 70% nodal measurement percentages in the IEEE 39-Bus with 0.5% noise level.

| % of Line measurements | 100% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|-------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                     | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0086                  | 0.0086 | 0.0087 | 0.0087 | 0.0085 | 0.0087 | 0.0085 | 0.0086 |
| GCNN-nodal             | 0.0213                  | 0.0211 | 0.0213 | 0.0207 | 0.0212 | 0.0209 | 0.0211 | 0.0210 |
| EdNdGCNN-hodgeLap      | 0.0212                  | 0.0210 | 0.0213 | 0.0208 | 0.0207 | 0.0208 | 0.0208 | 0.0209 |
| EdNdGCNN-edgeLap       | 0.0212                  | 0.0209 | 0.0213 | 0.0202 | 0.0203 | 0.0195 | 0.0198 | 0.0204 |

| % of Line measurements | 90% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                    | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0108                 | 0.0114 | 0.0114 | 0.0148 | 0.0117 | 0.0109 | 0.0114 | 0.0115 |
| GCNN-nodal             | 0.0230                 | 0.0236 | 0.0229 | 0.0234 | 0.0226 | 0.0229 | 0.0228 | 0.0228 |
| EdNdGCNN-hodgeLap      | 0.0226                 | 0.0229 | 0.0227 | 0.0232 | 0.0217 | 0.0224 | 0.0228 | 0.0226 |
| EdNdGCNN-edgeLap       | 0.0223                 | 0.0227 | 0.0223 | 0.0228 | 0.0208 | 0.0218 | 0.0220 | 0.0218 |

| % of Line measurements | 80% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                    | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0156                 | 0.0211 | 0.0129 | 0.0176 | 0.0152 | 0.0149 | 0.0140 | 0.0135 |
| GCNN-nodal             | 0.0236                 | 0.0238 | 0.0237 | 0.0235 | 0.0241 | 0.0241 | 0.0244 | 0.0240 |
| EdNdGCNN-hodgeLap      | 0.0230                 | 0.0235 | 0.0235 | 0.0230 | 0.0237 | 0.0231 | 0.0242 | 0.0239 |
| EdNdGCNN-edgeLap       | 0.0225                 | 0.0231 | 0.0233 | 0.0209 | 0.0224 | 0.0229 | 0.0234 | 0.0231 |

| % of Line measurements | 70% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                    | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0151                 | 0.0166 | 0.0165 | 0.0160 | 0.0196 | 0.0153 | 0.0153 | 0.0153 |
| GCNN-nodal             | 0.0249                 | 0.0248 | 0.0247 | 0.0254 | 0.0256 | 0.0250 | 0.0253 | 0.0254 |
| EdNdGCNN-hodgeLap      | 0.0234                 | 0.0242 | 0.0239 | 0.0238 | 0.0252 | 0.0243 | 0.0248 | 0.0251 |
| EdNdGCNN-edgeLap       | 0.0229                 | 0.0241 | 0.0237 | 0.0223 | 0.0242 | 0.0230 | 0.0238 | 0.0239 |

**Table A.2**

Evaluation of GAEN model using RMSE metric across various ranges of line and nodal measurements in the IEEE 39-bus scenario at 5% noise level.

| % of Line measurements | 100% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|-------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                     | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0111                  | 0.0111 | 0.0108 | 0.0111 | 0.0109 | 0.0110 | 0.0110 | 0.0109 |
| GCNN-nodal             | 0.0222                  | 0.0216 | 0.0215 | 0.0218 | 0.0216 | 0.0214 | 0.0211 | 0.0214 |
| EdNdGCNN-hodgeLap      | 0.0220                  | 0.0214 | 0.0213 | 0.0219 | 0.0217 | 0.0212 | 0.0212 | 0.0212 |
| EdNdGCNN-edgeLap       | 0.0219                  | 0.0211 | 0.0205 | 0.0219 | 0.0205 | 0.0203 | 0.0205 | 0.0208 |

| % of Line measurements | 90% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                    | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0125                 | 0.0127 | 0.0128 | 0.0124 | 0.0137 | 0.0133 | 0.0125 | 0.0123 |
| GCNN-nodal             | 0.0239                 | 0.0233 | 0.0230 | 0.0228 | 0.0233 | 0.0236 | 0.0231 | 0.0235 |
| EdNdGCNN-hodgeLap      | 0.0230                 | 0.0236 | 0.0233 | 0.0230 | 0.0221 | 0.0226 | 0.0230 | 0.0232 |
| EdNdGCNN-edgeLap       | 0.0227                 | 0.0235 | 0.0230 | 0.0231 | 0.0196 | 0.0220 | 0.0226 | 0.0219 |

| % of Line measurements | 80% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                    | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0140                 | 0.0139 | 0.0142 | 0.0153 | 0.0162 | 0.0153 | 0.0165 | 0.0164 |
| GCNN-nodal             | 0.0247                 | 0.0242 | 0.0247 | 0.0237 | 0.0246 | 0.0243 | 0.0244 | 0.0244 |
| EdNdGCNN-hodgeLap      | 0.0236                 | 0.0235 | 0.0240 | 0.0236 | 0.0242 | 0.0232 | 0.0238 | 0.0243 |
| EdNdGCNN-edgeLap       | 0.0230                 | 0.0234 | 0.0229 | 0.0226 | 0.0234 | 0.0228 | 0.0228 | 0.0236 |

| % of Line measurements | 70% Nodal measurements |        |        |        |        |        |        |        |
|------------------------|------------------------|--------|--------|--------|--------|--------|--------|--------|
|                        | 100                    | 90     | 80     | 70     | 60     | 50     | 40     | 30     |
| GAEN                   | 0.0169                 | 0.0179 | 0.0167 | 0.0171 | 0.0173 | 0.0167 | 0.0161 | 0.0168 |
| GCNN-nodal             | 0.0252                 | 0.0251 | 0.0255 | 0.0248 | 0.0251 | 0.0260 | 0.0250 | 0.0258 |
| EdNdGCNN-hodgeLap      | 0.0237                 | 0.0245 | 0.0244 | 0.0241 | 0.0239 | 0.0251 | 0.0247 | 0.0253 |
| EdNdGCNN-edgeLap       | 0.0230                 | 0.0237 | 0.0231 | 0.0235 | 0.0232 | 0.0243 | 0.0244 | 0.0243 |

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